Horizontal-Parallel ADC-less Sparsity-Clock-Aware RRAM CIM Macro for edge AI devices

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2026 IEEE International Symposium on Circuits and Systems (ISCAS) | 979-8-3315-7769-8/26/$31.00 ©2026 IEEE | DOI: 10.1109/ISCAS66217.2026.11562714

***Abstract*—The MAC operations essential to DNNs face significant memory wall challenges, motivating the development of Compute-in-Memory (CIM) as a promising solution. RRAM, being an emerging non-volatile memory technology, is well-suited for CIM implementations. However, existing RRAM-based CIM architectures suffer from two main limitations: (1) sensing margin (SM) degradation caused by simultaneously activating multiple vertical columns, along with the power and area overhead associated with ADCs; (2) degradation in inference accuracy under coarse-grained sparsity, and excessive control circuit costs for fine-grained sparsity handling. To address these issues, this paper presents a Horizontal-Parallel ADC-less Sparsity-Clock-Aware RRAM CIM Macro for edge AI devices. The design incorporates HPACM to enhance SM while eliminating ADCs, and employs SCAFA (Sparsity-Clock-Aware Fine-grained Architecture) to achieve full data-path fine-grained sparsity detection with minimal circuit and power overhead. Designed under 55nm CMOS technology, the macro achieves an energy efficiency of 7.37 TOPS/W@8b/8b and a peak throughput of 3.3 GOPS with a 92.25% inference accuracy.**

***Index Terms*—RRAM, Sparsity-Aware, Compute-In-Memory, Fine-Grained Sparsity, ADC-less**

I. Introduction

With the rapid development of AI technologies, deep learning models based on deep neural networks (DNNs) have demonstrated outstanding performance in various applications, including image recognition, large language models, and speech generation [1]–[3]. DNN inference and training are dominated by multiply-and-accumulate (MAC) operations, which are both computation- and data-intensive [4]. As AI models continue to scale in both dataset size and computational demand, conventional CMOS-based AI accelerators suffer from the memory-wall problem when processing MAC operations [5], [6]. Meanwhile, edge-AI systems such as those in automotive, IoT, and wearable devices impose stringent requirements on energy efficiency, standby power, wake-up latency, and silicon area [7].

To address these challenges, Compute-in-Memory (CIM) architectures have been proposed to eliminate frequent data movement and improve energy efficiency. Among various memory technologies, Resistive RAM (RRAM) is particularly suitable for CIM designs in edge AI devices, owing to its non-volatility, refresh-free operation, small cell area [8], [9], high ON/OFF ratio [10], [11], and excellent compatibility with CMOS technology [12], [13].

Although RRAM has been extensively studied in CIM [14]–[16].its application in CIM, faces several challenges: (1) Conventional RRAM-based CIM architectures perform in memory computations by simultaneously activating multiple rows along the vertical direction and aggregating analog signals through BL/SL before conversion via ADCs. This approach reduces energy consumption by shifting digital additions to the analog domain. However, it suffers from a trade-off between sensing margin (SM) and read power due to vertical parallelism, along with area and power overheads introduced by multi-bit ADCs, which ultimately restrict both parallelism and energy efficiency [14], [17].(2) Current CIM architectures aiming to exploit the high input sparsity inherent in DNN computations for acceleration and energy efficiency typically rely on coarse-grained block- or channel-wise skipping, or introduce large scale additional sparsity-encoding bits and control circuits for fine-grained skipping. The former provides optimization only for specific network models and may degrade inference accuracy, while the latter incurs extra circuit overhead in both power and area. [18]–[20].

To overcome these challenges, we propose a Horizontal-Parallel ADC-less Sparsity-Clock-Aware RRAM CIM Macro for edge AI devices, which employs two key innovations: (1) A Horizontal-Parallel Computing Mechanism (H-PACM) to performs computation by simultaneously activating multiple columns along the horizontal direction and directly converting signals into the digital domain through sense amplifiers (SAs). Leveraging an energy-efficient Sparsity-Clock Shifter Adder (S-CSA) for subsequent operations, it achieves high SM under low read power consumption, while avoiding the area and power overhead associated with ADCs. (2) A Sparsity-Clock-Aware Fine-grained Acceleration (SCAFA) scheme, which enables fine-grained sparsity awareness across the full datapath of the macro without introducing large scale additional encoding bits or control circuits. This approach effectively leverages unstructured sparsity to enhance both energy efficiency and throughput, while maintaining inference accuracy.The proposed macro, designed under 55nm CMOS, achieves an energy efficiency of 7.37 TOPS/W @8b/8b.

II. Structure of the Proposed RRAM CIM Macro

As shown in Fig. 1, the Horizontal-Parallel ReRAM CIM Architecture is composed of four 4Kb ReRAM subarrays. Each subarray is built around a 4Kb RRAM array, with the

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WL path consisting of a Row Decoder, Level Shifter, and WL Driver. Eight inputs are encoded by the Input Encoder and then fed into the subarray bit-serially through the SLs in parallel. The readout part of the array consists of the Horizontal-Parallel Readout Unit (H-PRU), Sparsity-Aware SA Gating Units (S-ASGU), Sparsity-Clock Shifter Adder, and Final Adder Tree. Among them, the HPRU together with the corresponding weight mapping scheme form the HPACM, while the S-ASGU, Sparsity-Clock Shifter Adder (S-CSA), and Final Adder Tree constitute the SCAFA mechanism.

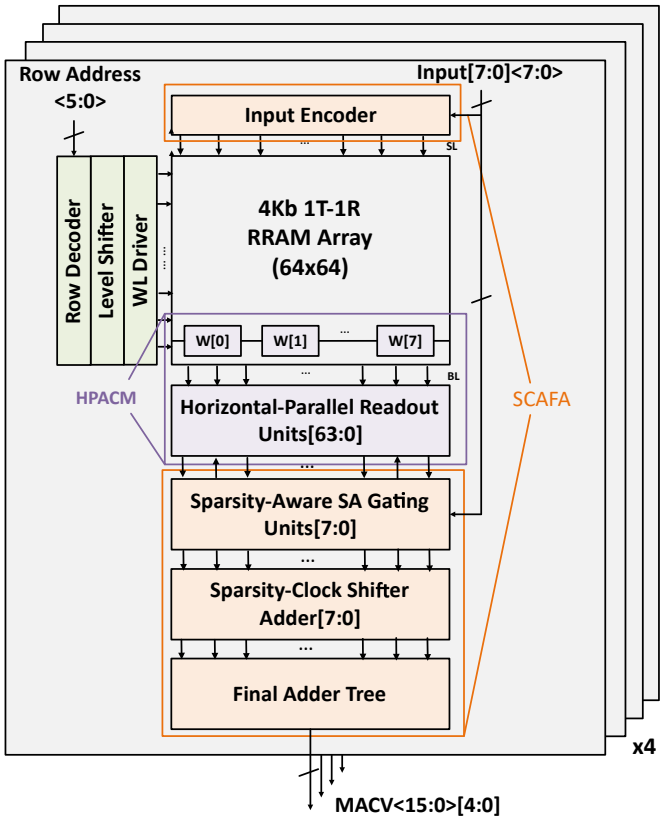


Fig. 1: Architecture of proposed CNM macro and the structure of 16Kb array.

III. The Proposed Horizontal-Parallel ADC-less Computing Mechanism

As shown in Fig. 2(a), the weight mapping scheme of HPACM and schematic of HPRU is illustrated. In each of the 64 rows within a subarray, the cells are evenly divided into eight groups, with each group consisting of eight RRAM cells. Each group represents an 8-bit weight, and each RRAM cell within the group corresponds to one bit of the weight. During MAC operations, in each cycle, one bit of the input corresponding to each weight group is applied through the SL to the subarray. The BL then develops an analog-domain voltage signal representing the dot-product result of IN[0]<k>*W[0]<7>, which is subsequently converted into digital result in the HPRU connected to the BL.

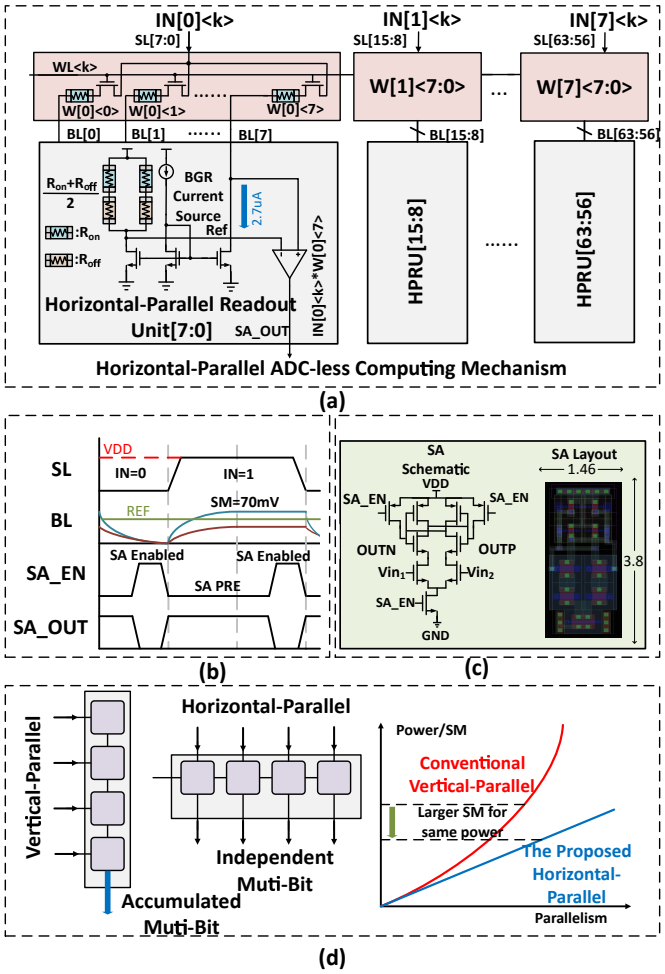


Fig. 2: (a) Structure of HPACM and HPRU (b) Waveform of HPRU. (c) Schematic and Layout of SA. (d) Comparison with conventional vertical-parallel strategy.

As shown in Fig. 2(b), during the dot-product operation, when the input is "1," the SL is pulled up to VDD, and the BL-side current-mirror transistor N1 generates a constant 2.7 μA current flowing from SL to GND. This produces two distinct BL voltages with a SM of about 70mV, depending on the resistance state of the RRAM cell. The SA then senses these voltages to generate a digital output. The bias voltage for the current mirror is provided by a band-gap reference current source, and the SA reference voltage is derived from a self-reference array to ensure process stability.

Due to the significant sparsity inherently present in input data, the array output contains substantial zeros. Conventional approaches that employ analog-domain aggregation followed by multi-bit ADC conversion thus incur considerable energy and area overhead. To achieve fine-grained sparsity awareness while avoiding ADC-induced redundancy, HPACM adopts direct sensing through SAs, followed by shift-accumulation in the digital domain using the energy-efficient S-CSA. The SA structure, depicted in Fig. 2(c), benefits from the SM enabled by horizontal parallelism, allowing the use of smaller

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transistors. This design occupies a compact layout area of 1.4 × 3.8 μm², resulting in reduced area and power consumption.

As illustrated in Fig. 2(d), unlike vertical parallelism, the output channels of individual cells in horizontal parallelism remain mutually independent. This architecture effectively mitigates the SM degradation caused by signal blending in conventional approaches. Consequently, it enables either a higher SM under the same read power budget or the same SM at a lower read power consumption.

IV. The Proposed Sparsity-Clock-Aware Fine-grained Acceleration

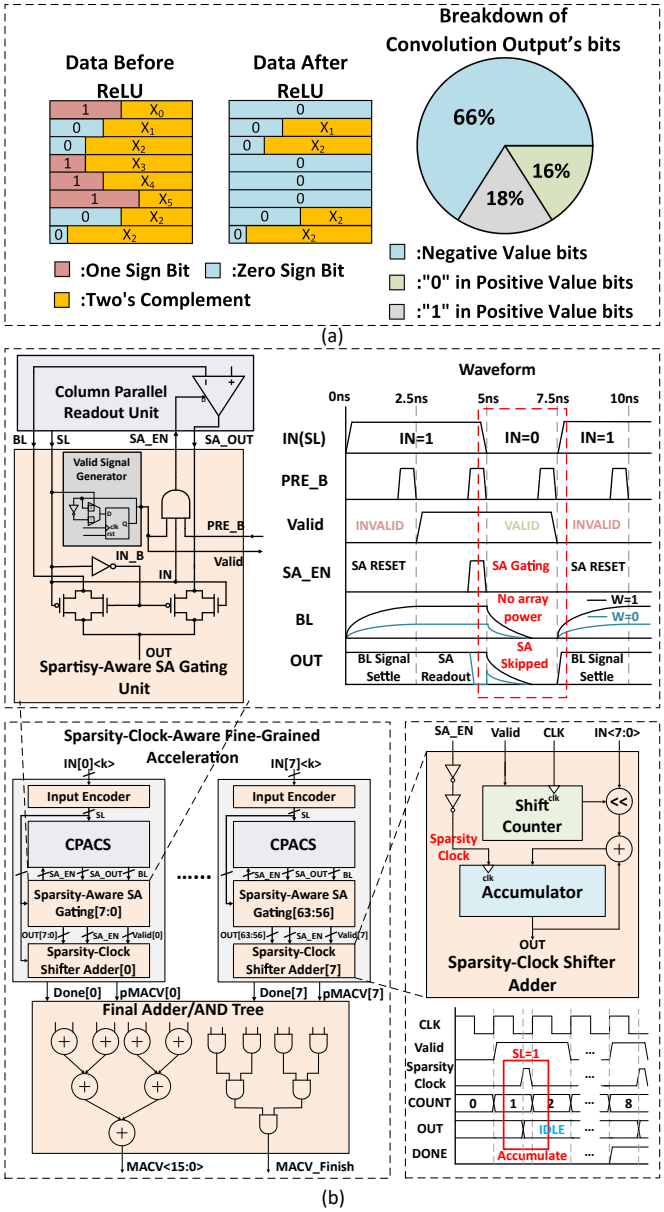


Fig. 3: (a) Analysis of input sparsities in DNN. (b) Overall structure of SCAFA.

As shown in Fig. 3(a), the sparsity of inputs in DNN MAC operations is analyzed. Due to the presence of ReLU functions between consecutive convolutional layers, the output of the previous layer—characterized by balanced positive and negative values—becomes highly sparse after the ReLU operation, as a large number of negative values are converted to zeros while positive numbers retain their sign bits. This results in substantial unstructured sparsity in the input, which cannot be effectively exploited by coarse-grained skipping methods such as channel- or block-level skipping. To address this, we propose a SCAFA mechanism capable of fine-grained sparsity detection and utilization within the datapath.

As illustrated in Fig. 3(b), the proposed SCAFA consists of Input Encoder, S-ASGU, Sparsity-Clock Shifter Adder (S-CSA).The structure and waveforms of S-ASGU are shown in Fig. 3(b). The S-ASGU consists of a Valid Signal Generator which consists of a DFF, a MUX, and a inverter generating the Valid signal, a 3-input AND gate for producing the SA_EN signal, and two transmission gates for output selection. The Valid signal indicates whether the output of the S-ASGU is valid in the current cycle, while SA_EN serves as the enable signal for the SA in the HPRU. The output transmission gates select either the SA_OUT from the HPRU when the input is "1" or the BL when the input is "0."

As illustrated in the waveform diagram, the input bits encoded by the Input Encoder are sequentially applied to the SL, where a "1" is held for two cycles and a "0" for one cycle. When the input is "1," the BL signal settles in the first cycle, with both Valid and SA_EN kept low; in the second cycle, the BL signal is fully established, SA_EN is asserted to perform readout, the Valid signal is raised, and the output is switched to SA_OUT. Conversely, when the input is "0," the BL quickly discharges to GND within a single cycle, current-mirror transistor is automatically shutdown to save energy, while the , SA_EN remains low so the SA is disabled, the Valid signal is raised, and the output is switched to BL, producing a digital "0."

The output from the SAGUs is fed into the S-CSA for shift-and-accumulate operations. The S-CSA consists of a shift counter and an accumulator: the former is responsible for counting the number of processed inputs and their corresponding shift amounts, while the latter accumulates outputs across different cycles. Since shift-accumulate operations contribute significantly to power consumption—largely dominated by dynamic power in the clock network—the S-CSA utilizes the SA_EN signal as its Sparsity Clock, leveraging input sparsity to reduce clock switching frequency and thereby lower power consumption. When the shift counter reaches a value of 8, the DONE[k] signal is asserted, and the S-CSA outputs pMACV[k]. Subsequently, the eight channels of pMACV[7:0] and DONE[7:0] are processed by a Final Adder Tree to generate MACV and MAC_Finish signals.

The SCAFA achieves fine-grained sparsity utilization across the entire datapath—including the array, SAs, and S-C Shifter—without introducing additional sparsity encoding bits or large control circuitry. With only a few extra AND gates and register bits, it maximizes power reduction by fully exploiting input sparsity throughout the computation flow. Meanwhile,

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the processing time for sparse inputs is compressed to half that of non-sparse inputs, thereby improving overall macro throughput.

V. Simulation Results and Application in DNN

To evaluate the process stability of HPACM, we measured Offset voltage of SA and examined the impact of RRAM device variability on computation accuracy. As shown in Fig.4 (a), the Monte-Carlo simulation of SA is illustrated, with a Voffset about 50mV.As depicted in Fig. 4(b), the RRAM TMR was reduced from 15 to 4, and for each case we measured the BL-side SM as well as the readout accuracy of the SA under Monte Carlo simulation. The results show that the SM decreases from 71 mV to 40 mV and the SA readout accuracy drops from 100 % to 99.8 %.

To verify the energy-efficiency improvement brought by the SCAFA mechanism, Fig. 4(c) presents the measured energy efficiency under different input sparsities, comparing macro with and without SCAFA. In realistic DNN scenarios characterized by high input sparsity, the SCAFA mechanism achieves more than a 3x improvement in energy efficiency, raising the system's performance from 1.46 TOPS/W to 7.44 TOPS/W. Fig. 4(d) shows the power breakdown of the macro, in which the power of circuit additionally introduced by SCAFA accounts for only 2%.

As shown in Fig. 4(e), the timing diagram illustrates the MAC operation of the marco. After the MAC_Start signal is asserted, the input bits are sequentially applied to the SL line on a cycle-by-cycle basis. When the final input bit has been processed, according to SCAFA, DONE[7:0] is asserted, so the MAC_Finish signal goes high and MACV outputs the complete MAC result.

The proposed macro is applied to an INT8-quantized AlexNet on the CIFAR-10 dataset. As illustrated in Fig. 4(f), which depicts the practical DNN application scenarios, the DNN consists of multiple convolution-ReLU layers. Simulation results show that the macro achieves a top-1 inference accuracy of 92.25%, with an accuracy loss of only 0.57%, an energy efficiency of 7.44 TOPS/W @8b/8b, and a peak throughput of 3.3 GOPS.

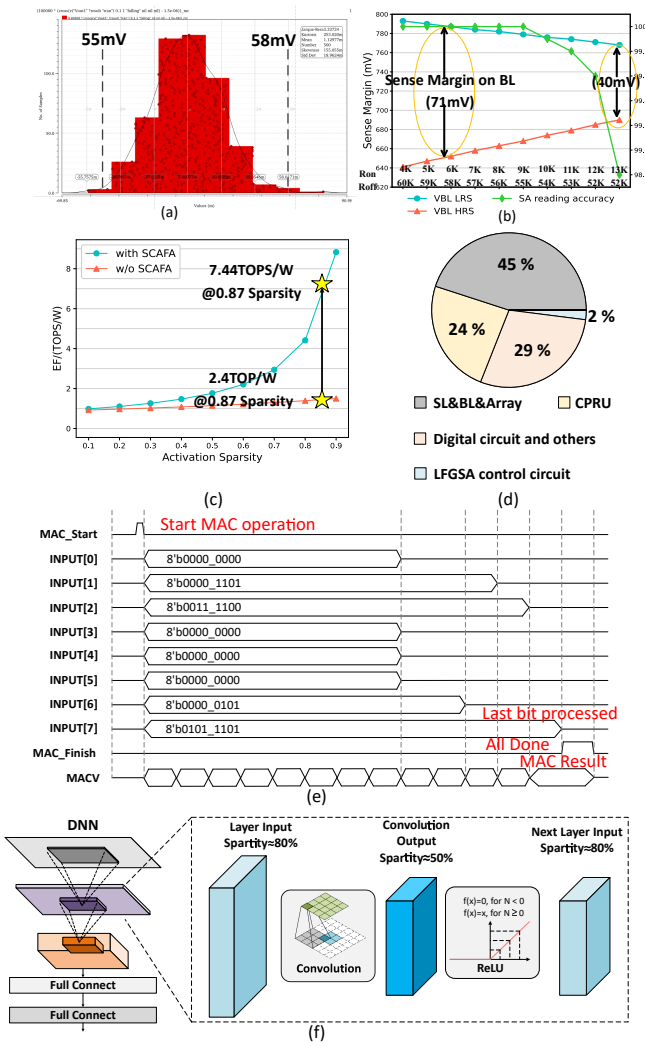


Fig. 4: (a) Offset voltage of SA. (b) Process stability simulation of HPACM. (c)Energy efficiency under different sparsity with/without LFGSA. (d) Power breakdown of the marco (e) Timing diagram of MAC operation. (f) Practical DNN application scenarios of the proposed macro.

VI. CONCLUSION

This work proposes a Horizontal-Parallel ADC-less Sparsity-Clock-Aware RRAM CIM Macro for edge AI devices. The design incorporates the HPACM, achieving a 70 mV sensing margin under a 2.7 µA read current, while replacing ADCs with compact SAs to ensure readout accuracy despite large RRAM resistance variations and process fluctuations. Moreover, the proposed design employs the SCAFA mechanism, which achieves full-datapath high-efficiency sparsity utilization by only introducing little extra power and circuit. Under highly sparse activation scenarios in practical DNNs, the macro achieves a 5× energy-efficiency improvement and a peak throughput of 3.3 GOPS. When applied to AlexNet on the CIFAR-10 dataset, the proposed macro attains an energy efficiency of 7.37 TOPS/W @ 8b/8b.

TABLE I: **Comparison with other CIM designs**

	JSSC2022 [18]	VLSI2023 [17]	ISCAS2025 [19]	JSSC2023 [20]	This Work
Technology	65nm	28nm	28nm	28nm	55nm
Cell Type	SRAM+ RRAM	RRAM	RRAM	SRAM	RRAM
Accumulation Direction	Vertical	Horizontal	Vertical	Vertical	Horizontal
Supply Voltage	0.9V	0.8V	0.9V	0.65V	1.2V
Sparsity Type	block-wise	–	fine-grained	block-wise	fine-grained
Precision	INT4	INT8	INT8	INT8	INT8
Accuracy	91.74%	–	–	79.28%	92.25%
Normalized EF (TOPS/W)[1]	5.08	6.15	6.5	7.35[2]	7.37
FoM[3]	4.66	-	-	5.75	6.79

1: Normalized Energy Efficiency = 1/(Technology*VDD²*Input_Precision*Output_Precision).
2: Under the same sparsity.
3:FoM = Normalized EF * Inference accuracy.

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